

PRE UNIT TEST

Testing for Ions, Water, Flame Tests, Precipitation and Limiting Reactants

Guided examination paper

Student name:	Class:
Teacher:	Date:
Time allowed: 60 minutes	Total marks: 40

Exam instructions

Answer all questions. Show all working in calculations. Use appropriate chemical formulae, state symbols and units. For practical questions, distinguish clearly between the reagent used, the observation made and the conclusion drawn.

Paper structure

Section	Question style	Marks
A	Multiple choice	8
B	Structured free response	12
C	Practical analysis	10
D	Limiting reactants calculation	10

SECTION A - MULTIPLE CHOICE

8 marks

Guidance

Select one answer for each question. Write the letter A, B, C or D on the answer line.

1. Which flame colour indicates the presence of potassium ions? [1]

A red	B yellow
C lilac	D blue-green

2. Which substance is used to test chemically for the presence of water? [1]

A anhydrous copper(II) sulfate	B aqueous sodium hydroxide
C acidified silver nitrate	D limewater

3. A white precipitate forms when acidified silver nitrate is added to a solution. Which ion is present? [1]

A bromide	B chloride
C iodide	D sulfate

4. Which observation is expected when aqueous sodium hydroxide is added to a solution containing Cu^{2+} ions? [1]

A green precipitate	B white precipitate
C light-blue precipitate	D red-brown precipitate

5. Which reagent is used to test for sulfate ions? [1]

A dilute hydrochloric acid only	B acidified barium nitrate solution
C aqueous sodium hydroxide	D acidified silver nitrate solution

6. What is a precipitate? [1]

A a gas produced in a reaction	B an insoluble solid formed from solutions
C a soluble salt dissolved in water	D a colour change in a flame

7. The limiting reactant is the reactant that... [1]

A has the greatest mass	B has the highest concentration
C is completely used up first	D is always a solid

8. Which ionic equation represents the formation of silver chloride? [1]

A $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$	B $\text{Ag}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{AgCl}(\text{aq})$
C $\text{AgNO}_3(\text{aq}) \rightarrow \text{Ag}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$	D $\text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{NaCl}(\text{s})$

SECTION B - MULTIPLE CHOICE

12 marks

9 Testing water and assessing purity**[4]**

(a) Describe the chemical test for water using anhydrous copper(II) sulfate. Include the starting colour and the positive result. **[2]**

(b) A liquid boils sharply at 100 °C at standard atmospheric pressure. Explain what this suggests about the purity of the water. **[2]**

10 Carrying out a flame test**[4]**

(a) Describe a safe method for carrying out a flame test on a solid salt. Your method should include how the wire loop is cleaned. **[3]**

(b) Explain why the wire loop must be cleaned before testing a new sample. **[1]**

11 Precipitation and ionic equations**[4]****Guidance**

A strong answer names the reagent, gives a precise observation and then identifies the ion.

- (a) Describe how to test an aqueous sample for chloride ions. State the reagent(s) and the positive observation. **[2]**

- (b) Write the ionic equation, including state symbols, for the formation of silver chloride. **[2]**

SECTION C - PRACTICAL ANALYSIS

10 marks

12 Identifying two unknown solutions

[10]

A student carries out tests on two unknown solutions, A and B. The results are shown in the table.

Solution	Test	Observation	Inference to be completed
A	Flame test	lilac flame	
A	Acidify, then add aqueous silver nitrate	white precipitate	
B	Add aqueous sodium hydroxide	green precipitate; surface slowly turns brown	
B	Acidify, then add aqueous barium nitrate	white precipitate	

(a) Identify the positive ion and negative ion present in solution A. [2]

(b) Give the formula of the dissolved salt in solution A. [1]

(c) Identify the positive ion and negative ion present in solution B. [2]

(d) Give the formula of the dissolved salt in solution B. [1]

(e) Write the ionic equation, including state symbols, for the sulfate precipitation test. [2]

(f) State one safety precaution for this investigation and explain why it is needed. [2]

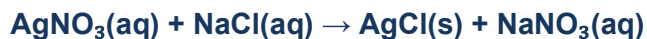
SECTION D - LIMITING REACTANTS

10 marks

13 Formation of a silver chloride precipitate

[10]

25.0 cm³ of 0.200 mol dm⁻³ silver nitrate solution is mixed with 20.0 cm³ of 0.150 mol dm⁻³ sodium chloride solution.

**Calculation guidance**

Convert cm³ to dm³ before using $n = cV$. Compare reacting amounts using the mole ratio in the balanced equation. Use $m = nM_r$ for the precipitate mass.

(a) Calculate the amount, in moles, of AgNO₃ used. [1]

(b) Calculate the amount, in moles, of NaCl used. [1]

(c) Identify the limiting reactant. Justify your answer using the mole ratio. [1]

(d) Determine the amount, in moles, of AgCl formed. [1]

(e) Calculate the maximum mass of AgCl precipitate formed. Give your answer to three significant figures.
[$M_r(\text{AgCl}) = 143.5$] [2]

(f) Calculate the amount, in moles, of the reactant left in excess after the reaction is complete. [2]

(g) Calculate the concentration of the excess reactant remaining, in mol dm^{-3} . [2]

TOTAL FOR PAPER = 40 MARKS

STUDENT REVIEW**complete after marking****Reflection**

Use the mark scheme and teacher feedback to identify the next step that will improve your examination performance.

Skill / topic	Confident	Developing	Action needed
Flame-test colours and procedure	☐	☐	
Tests for cations and anions	☐	☐	
Testing for water and purity	☐	☐	
Recording practical observations	☐	☐	
Writing ionic equations	☐	☐	
Limiting-reactant calculations	☐	☐	

My priority revision target:

Group																			
1	2											13	14	15	16	17	18		
atomic number atomic symbol name relative atomic mass Key H hydrogen 1.0																		2	
3 Li lithium 6.9	4 Be beryllium 9.0											5 B boron 10.8	6 C carbon 12.0	7 N nitrogen 14.0	8 O oxygen 16.0	9 F fluorine 19.0	10 Ne neon 20.2		
11 Na sodium 23.0	12 Mg magnesium 24.3											13 Al aluminium 27.0	14 Si silicon 28.1	15 P phosphorus 31.0	16 S sulfur 32.1	17 Cl chlorine 35.5	18 Ar argon 39.9		
19 K potassium 39.1	20 Ca calcium 40.1	21 Sc scandium 45.0	22 Ti titanium 47.9	23 V vanadium 50.9	24 Cr chromium 52.0	25 Mn manganese 54.9	26 Fe iron 55.8	27 Co cobalt 58.9	28 Ni nickel 58.7	29 Cu copper 63.5	30 Zn zinc 65.4	31 Ga gallium 69.7	32 Ge germanium 72.6	33 As arsenic 74.9	34 Se selenium 79.0	35 Br bromine 79.9	36 Kr krypton 83.8		
37 Rb rubidium 85.5	38 Sr strontium 87.6	39 Y yttrium 88.9	40 Zr zirconium 91.2	41 Nb niobium 92.9	42 Mo molybdenum 95.9	43 Tc technetium –	44 Ru ruthenium 101.1	45 Rh rhodium 102.9	46 Pd palladium 106.4	47 Ag silver 107.9	48 Cd cadmium 112.4	49 In indium 114.8	50 Sn tin 118.7	51 Sb antimony 121.8	52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3		
55 Cs caesium 132.9	56 Ba barium 137.3	57–71 lanthanoids		72 Hf hafnium 178.5	73 Ta tantalum 180.9	74 W tungsten 183.8	75 Re rhenium 186.2	76 Os osmium 190.2	77 Ir iridium 192.2	78 Pt platinum 195.1	79 Au gold 197.0	80 Hg mercury 200.6	81 Tl thallium 204.4	82 Pb lead 207.2	83 Bi bismuth 209.0	84 Po polonium –	85 At astatine –	86 Rn radon –	
87 Fr francium –	88 Ra radium –	89–103 actinoids		104 Rf rutherfordium –	105 Db dubnium –	106 Sg seaborgium –	107 Bh bohrium –	108 Hs hassium –	109 Mt meitnerium –	110 Ds darmstadtium –	111 Rg roentgenium –	112 Cn copernicium –	114 flerovium –					116 Lv livermorium –	
lanthanoids																			
		57 La lanthanum 138.9	58 Ce cerium 140.1	59 Pr praseodymium 140.9	60 Nd neodymium 144.4	61 Pm promethium –	62 Sm samarium 150.4	63 Eu europium 152.0	64 Gd gadolinium 157.3	65 Tb terbium 158.9	66 Dy dysprosium 162.5	67 Ho holmium 164.9	68 Er erbium 167.3	69 Tm thulium 168.9	70 Yb ytterbium 173.1	71 Lu lutetium 175.0			
		actinoids																	
		89 Ac actinium –	90 Th thorium 232.0	91 Pa protactinium 231.0	92 U uranium 238.0	93 Np neptunium –	94 Pu plutonium –	95 Am americium –	96 Cm curium –	97 Bk berkelium –	98 Cf californium –	99 Es einsteinium –	100 Fm fermium –	101 Md mendelevium –	102 No nobelium –	103 Lr lawrencium –			